

Lecture 1: Introduction to the Course and Preliminaries

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Course Description

- The purpose of this course is to introduce the foundational tools of microeconomic analysis.
- The overarching topics in this course are how consumers make choices both under certainty and uncertainty; how firms take decisions subject to technological constraints and in relation to the market structure they operate in and the organization of markets in the presence of complete and asymmetric information.
- The real world is very complex, so we use **models** to gain understanding.
- This class will introduce the basic models that populate economic analysis.
- We shift the focus from Principles of Economics, from learning facts to developing techniques for analyzing economic questions.

Course Requirements

- **Principles of Economics:** ECO 108 at Stony Brook.
- A math course at the calculus level or higher. This would be MAT122, MAT123, or AMS151.
- More precisely, the requirement is a course on **single variable calculus**.
- You must be familiar with concepts such as functions, continuity and derivatives.
- This week's recitation covers precisely that to freshen up your memory.
- If you find that you don't recall any of this material, talk to me ASAP!
- We will develop new math tools as needed during the course.

Course Materials

- **Required!** *Microeconomic Theory: Basic Principles and Extensions* by Walter Nicholson and Christopher Snyder, 11th edition, Cengage Learning, 2011.
- *Intermediate Microeconomics with Calculus* by Hal R. Varian, 1st edition, W. W. Norton & Company, 2014.
- *Advanced Microeconomic Theory* by Geoffrey A. Jehle and Philip J. Reny, 3rd edition, Pearson, 2011.
- *Recommended. Essential Mathematics for Economic Analysis* by K. Sydsaeter and P. Hammond, Prentice-Hall 2008.
- I will assign some recent research articles and other readings to illustrate the models that we will cover in class and to provide real world applications. This material and my lecture slides will be posted on BlackBoard so it is your responsibility to check it regularly.

Keeping up with the course

- Although we will move at a reasonable pace, the course is demanding.
- Attending recitation sections is **required**. You will discuss past problem sets and elaborate on the material discussed in the lecture.
- Attending lectures, **also highly recommended!**
- Doing the readings from the book is also **required**.
- Getting help: Talk to me after class, during office hours, go to recitations, attend the TA's office hours and also the undergraduate TA's office hours.

Grading Policy

- Midterms (60%)
- Problem Sets (40%)

A letter grade based on the sum of the components above will be assigned as follows:

Letter Grade	Score Achieved
A	[93, 100]
A–	[90, 93)
B+	[87, 90)
B	[83, 87)
B–	[80, 83)
C+	[77, 80)
C	[73, 77)
C–	[70, 73)
D+	[67, 70)
D	[60, 67)
F	[0, 60)

Problem sets and Exams

- 10–12 Problem Sets during the course. I will announce in class and in BlackBoard when a new problem set is posted. Problems sets are due *one week* after they are announced and you must deliver them at the end of the class.
 - You will four exams, three midterms and one final exam. Each midterm is non-cumulative and the final is cumulative, covering everything we will see in the course.
 - I will drop the lowest scoring exam to compute your exam average.
- No need to take the final if you are happy with the midterms!

What is Economics?

- An exciting time to study Economics!
- We faced the greatest crisis since the Great Depression.
- \$787 billion has been pumped into the economy in the form of a stimulus package by the US government.
- \$700 billion has been spent on the Troubled Asset Relief Program for Banks.
- During the aftermath of the crisis, unemployment reached very high levels, remained high for a long period of time, and just recently started to reach pre-crisis levels.
- Big debates going on on health care reform, government deficits, climate change, immigration policy, etc.
- We need answers to these and many other questions!
- All of these questions come under the purview of the discipline of Economics (as also other fields of study).

So, what is Economics?

- Broadly defined, is the study of how to allocate resources when they are scarce and we have unlimited wants.
- ☺ Economics is what economists do!
- In terms of methodology, Economics is characterized by the use of Optimization Theory, Statistics, Game Theory, Decision Science, and recently even Computer Science.
- By subject of study: explain human behavior in several contexts, humans as consumers and decision makers in firms or in families, as politicians, etc.
- Economics can be used to study: trade, taxes, growth, antitrust policy, crime, marriage, war, laws, media, corruption, addictive behavior, social networks, etc.
- Economists: people that try to explain these issues and others with the help of our way of thinking.
- The frontier of research is to push the frontier of economics.

What is Economics? On Building Theories

- What is a theory? A set of propositions about something.
- When can we claim that a theory is scientific and another is not scientific?
- Consider the following example: ‘The Moon is in Libra, making this an ideal time to express your opinions and develop your creative ideas.’ How could you show, even in principle, that this statement was false? Or how about this: ‘The Moon is in Libra, making this a really bad time to express your opinions and develop your creative ideas.’
- According to Karl Popper, we can’t ever aspire to show that a hypothesis is true, the most we can aspire is to be able to show that it is false. This is called **falsifiability**.
- Falsifiability is the property of a theory to be able to be shown false.
- You can think of falsifiability as being able to be tested.
- “The sun may or may not rise tomorrow.” If the sun rises tomorrow, then this is true. If the sun doesn’t rise, it is also true. Therefore you can’t empirically test this statement.

Using the Popperian tradition in economics

- Economist like to think that we try to adhere to Popperian methods.
- We move to stylized facts, to hypothesis formation to testing hypothesis.
- Remember, according to Popper, a good theory is one that can be proved false (we can never prove true a theory, but we can falsify it.)
- We seek to build theories about the world with sensible hypotheses that can be tested.
- Economic theory is the exercise in hypothesis formation using the language of mathematics to formalize assumptions about human behavior, market organization, distribution of information among individuals, etc.

On the use of models

- As said at the beginning, we use models to explain complicated phenomena from the real world.
- An usual jab at economics is that our models are too simplistic, that we leave too many things out.
- **Of course we do!** Otherwise it is not a good model!
- A good model should leave all the extraneous things out, leaving only the most relevant parts, so that a very clear argument can be made.
- It is useful to leave things out! Otherwise is too complicated.
“Everything should be made as simple as possible, but not simpler.”
- In any good fable, we take a road into the imaginary, while keeping some details true to reality. A good fable imparts an advice that is useful.
- Replace fable with economic model in the last sentence.
- Focusing on the most salient features of a phenomenon when building a model, helps keep things simple, so we can build clear arguments.

The assumptions of Economics

- We usually rely on optimizations assumptions as a core tool in economics.
- Agents in our models are assumed to pursue some goal, i.e. they are optimizing.
- This framework is easy to work with and permeates most of modern microeconomics.
- We have three core principles:
 1. People respond to incentives.
 2. An equilibrium concept that assumes the absence of free lunches.
 3. A welfare criterion that says that more choices are better.
- Milton Friedman pointed a distinction of the field:
 1. Positive economics: Why the world is the way it is and looks the way it does.
 2. Normative economics: How the world can be improved.

My Goals for the course

- You will understand basic microeconomic mechanisms, i.e. the actors that populate our economic fables and how this fables work.
- You will be able to evaluate and challenge economic arguments.
- Hope that you gain an appreciation of the economic way of looking at the world.

An example

You may recall, from the ECO108 class about the assumption that firms are profit maximizing.

- Suppose that a firm can sell all the output it wishes at price p .
- The cost of producing, depends on quantity produced q .
- Profits are given by

$$\pi(q) = pq - C(q) \quad (1)$$

- Notice the assumption: price taking behavior by the firm.
- Maximization of (1) entails taking the first order condition:

$$\frac{d\pi}{dq} = p - C'(q) = 0 \rightarrow p = C'(q^*) \quad (2)$$

- This means, the profit maximization is obtained when the firm puts **price equal to marginal cost**.

Example continued...

- The second order condition to check that indeed we have found a maximum is:

$$\frac{d^2\pi}{dq^2} = -C''(q) < 0 \rightarrow C''(q^*) > 0 \quad (3)$$

- Marginal cost should be increasing to get a maximum.
- How would the firm react to a change in price?

$$\frac{d[p - C'(q) = 0]}{dp} = 1 - C''(q^*) \cdot \frac{dq^*}{dp} = 0 \rightarrow \frac{dq^*}{dp} = \frac{1}{C''(q^*)} > 0 \quad (4)$$

- We have obtained a testable implication: other things equal, a profit maximizing firm should respond to a positive price change by increasing its output.

Another example, gains from trade, bilateral trade

- Seller, values a single homogeneous object at c .
- Buyer values the same object at v .
- Trade occurs at price p .
- Seller gets $p - c$, buyer gets $v - p$, no trade entails 0.
- Whenever $v > c$, there is possibility to gain from trade at some $p \in [c, v]$.
- How to determine the actual price paid?
- At any price at which trade is possible, we get an allocation in which the buyer gets the object and the seller gets the payment. This is **efficient**.
- An allocation is efficient (Pareto) if there is no other allocation that makes at least one agent strictly better off without making any other agent worse off.

Generalizing the example

- The framework can incorporate many buyers and sellers and allow for production.
- We are interested in not only how specific markets function, but also how should they be organized.
- NYSE, NASDAQ, E-bay and Google for example.

Example

Two types of cards (red and black) with numbers between 2 and 10 are handed out to the students. If the student receives a red card they are a seller, and the number reflects their cost. If the student receives a black card they are a buyer, and this reflects their valuation. The number on the card is private information. Trade then takes place according to the following three protocols.

1. **Bilateral Trading:** One seller and one buyer are matched before receiving their cards. The buyer and seller can only trade with the individual they are matched with. They have 5 minutes to make offers and counter offers and then agree (or not) on the price.
2. **Pit Market:** Buyer and seller cards are handed out to all students at the beginning. Buyers and sellers then have 5 minutes to find someone to trade with and agree on the price to trade.
3. **Double Auction:** Buyer and seller cards are handed out to all students at the beginning. The initial price is set at 6 (the middle valuation). All buyers and sellers who are willing to trade at this price can trade. If there is a surplus of sellers the price is decreased, and if there is a surplus of buyers then the price is increased. This continues for 5 minutes until there are no more trades taking place.